

Rectilinear Kinematics (Ch. 12)

$v = \frac{ds}{dt}$ – velocity is the time rate of change of position

$a = \frac{dv}{dt}$ – acceleration is the time rate of change of velocity

$a ds = v dv$ – eliminates time; use when a is a function of position

$v_{avg} = \frac{\Delta s}{\Delta t}$, $(v_{sp})_{avg} = \frac{s_T}{\Delta t}$ – average velocity uses displacement; average speed uses total distance s_T

Constant velocity ($a = 0$):

$s = s_0 + vt$ – uniform motion; s - t graph is a straight line of slope v

Constant acceleration ($a = a_c$) – pick the equation that leaves out the quantity you neither know nor want:

$v = v_0 + a_c t$ – no s ; velocity as a function of time

$s = s_0 + v_0 t + \frac{1}{2} a_c t^2$ – no v ; position as a function of time

$v^2 = v_0^2 + 2a_c(s - s_0)$ – no t ; velocity as a function of position

$s = s_0 + \frac{1}{2}(v_0 + v)t$ – no a_c

$s = s_0 + vt - \frac{1}{2} a_c t^2$ – no v_0

$v_{avg} = \frac{1}{2}(v_0 + v)$ – holds for constant a_c only

Free fall: $a_c = -g = -9.81 \text{ m/s}^2$ (+ y up) – $v = 0$ at maximum height; keep the sign convention consistent

s - t , v - t , a - t graphs:

v = slope of the s - t graph at that instant (ds/dt)

a = slope of the v - t graph at that instant (dv/dt)

Δv = area under the a - t graph (change in velocity)

Δs = area under the v - t graph (displacement)

Variable acceleration – which relation to use:

$a = f(t)$: $v = v_0 + \int_0^t a dt$, $s = s_0 + \int_0^t v dt$ – integrate directly

$a = f(v)$: $t = \int_{v_0}^v \frac{dv}{f(v)}$, $s - s_0 = \int_{v_0}^v \frac{v dv}{f(v)}$ – from $a = dv/dt$ and $a ds = v dv$, separate variables

$a = f(s)$: $v^2 = v_0^2 + 2 \int_{s_0}^s a ds$ – from $a ds = v dv$

$v = f(s)$: $a = v \frac{dv}{ds}$, $t = \int_{s_0}^s \frac{ds}{v}$ – differentiate for a , separate

$v = ds/dt$ for t

Direction reversal: solve $v = 0$ for the turning time and split the interval there – total distance $\neq$ displacement

Curvilinear & Projectile Motion (Ch. 12)

$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}}$ – $\mathbf{v}$ is always tangent to the path

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \ddot{\mathbf{r}}$ – $\mathbf{a}$ is generally NOT tangent to the path in curvilinear motion

$v_x = \dot{x}$, $v_y = \dot{y}$, $v_z = \dot{z}$ – rectangular velocity components; each is an independent rectilinear problem

$a_x = \ddot{x}$, $a_y = \ddot{y}$, $a_z = \ddot{z}$ – rectangular acceleration components; combine vectorially

$x = x_0 + (v_x)_0 t$

$y = y_0 + (v_y)_0 t - \frac{1}{2} g t^2 - a_x = 0$ (constant v_x), $a_y = -g$ ($g \approx 9.81 \text{ m/s}^2$)

$v_y = (v_y)_0 - gt$

$v_y^2 = (v_y)_0^2 - 2g(y - y_0)$ – solve x - and y -motion separately, linked by common time t

Projectile shortcuts (launch speed v_0 at angle θ_0 above horizontal):

$y = x \tan \theta_0 - \frac{gx^2}{2v_0^2 \cos^2 \theta_0}$ – trajectory from the launch point; eliminates t

$R = \frac{v_0^2 \sin 2\theta_0}{g}$, $t_f = \frac{2v_0 \sin \theta_0}{g}$ – range and time of flight; landing at launch height only

$H = \frac{v_0^2 \sin^2 \theta_0}{2g}$ – maximum height above launch point ($v_y = 0$)

$v = \sqrt{v_x^2 + v_y^2}$ – resultant speed

$\theta = \tan^{-1}(v_y/v_x)$ – measured from x -axis to velocity vector; same approach for acceleration vector

Normal-Tangential (n-t) Coordinates (Ch. 12)

$\mathbf{v} = \dot{s} \hat{\mathbf{u}}_t$ – velocity always tangent to path, direction of increasing s

$\mathbf{a} = a_t \hat{\mathbf{u}}_t + a_n \hat{\mathbf{u}}_n$ – acceleration splits into tangential and normal parts

$a_t = \dot{v} = v \frac{dv}{ds}$ – tangential component; changes the *speed*

$a_n = \frac{v^2}{\rho}$ – normal component; changes *direction*; always points toward the center of curvature

$a = \sqrt{a_t^2 + a_n^2}$ – resultant magnitude of $\mathbf{a}$

$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$ – radius of curvature, needed for a_n when path $y = f(x)$ is analytic

Arc length (path coordinate s):

$s = r\theta$ – circular arc, θ in radians; so $v = r\dot{\theta}$, $a_t = r\ddot{\theta}$, $a_n = r\dot{\theta}^2$

$s = \int_{x_1}^{x_2} \sqrt{1 + (dy/dx)^2} dx$ – path given as $y = f(x)$

$s = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + (dr/d\theta)^2} d\theta$ – path given as $r = f(\theta)$

$s = \int_{t_1}^{t_2} v dt = \int_{t_1}^{t_2} \sqrt{\dot{x}^2 + \dot{y}^2} dt$ – distance travelled along the path; v is speed

Circular path: $\rho = \text{const} = r$

Circular motion with constant angular acceleration ($\ddot{\theta} = \alpha_c$, $\omega = \dot{\theta}$):

$\omega = \omega_0 + \alpha_c t$

$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha_c t^2$

$\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$ – same form as constant a_c with $s \rightarrow \theta$, $v \rightarrow \omega$, $a \rightarrow \alpha$

Straight-line path: $\rho \rightarrow \infty \Rightarrow a_n = 0$, $a = a_t$ – n - t coordinates reduce to ordinary rectilinear motion

Polar & Cylindrical Coordinates (Ch. 12)

$\mathbf{r} = r\hat{\mathbf{u}}_r$ – position

$v_r = \dot{r}$ – radial velocity component

$v_\theta = r\dot{\theta}$ – transverse velocity component

$v = \sqrt{v_r^2 + v_\theta^2}$ – always tangent to the path, as it must be

$a_r = \ddot{r} - r\dot{\theta}^2$ – radial acceleration component

$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$ – transverse; the $2\dot{r}\dot{\theta}$ term is essential whenever r varies with time

$a = \sqrt{a_r^2 + a_\theta^2}$ – resultant magnitude

$\dot{r} = \frac{dr}{d\theta} \dot{\theta}$, $\ddot{r} = \frac{d^2r}{d\theta^2} \dot{\theta}^2 + \frac{dr}{d\theta} \ddot{\theta}$ – chain rule; use when the path is given as $r = f(\theta)$

$\tan \psi = \frac{r}{dr/d\theta}$ – ψ = angle from the extended radial line to the

tangent; locates the normal force direction (slotted arm, cam)

$\dot{u}_r = \dot{\theta} \hat{u}_\theta$, $\dot{u}_\theta = -\dot{\theta} \hat{u}_r$ – unit-vector rates; source of the extra terms above

$v_z = \dot{z}$, $a_z = \ddot{z}$ – cylindrical adds the z -axis; combine vectorially: $\mathbf{v} = v_r \hat{u}_r + v_\theta \hat{u}_\theta + v_z \hat{u}_z$

Circular path in polar form ($r = \text{const}$): $a_r = -r\dot{\theta}^2$, $a_\theta = r\ddot{\theta}$ – matches n - t with $\rho = r$

Dependent & Relative Motion (Ch. 12)

$s_A + l_{CD} + s_B = l_T$ – write a position-coordinate equation using the total constant cable length

$\dot{s}_A + \dot{s}_B = 0$ – differentiate once for velocities

$\ddot{s}_A + \ddot{s}_B = 0$ – differentiate again for accelerations; opposite signs mean the blocks move oppositely (coefficients = number of cable segments changing length at each moving pulley)

$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$ – relative position; translating, non-rotating $x'y'$ frame attached to A

$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$ – relative velocity

$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$ – relative acceleration; for two independently-moving particles (e.g. two aircraft, a boat crossing a current) – solve as a vector equation in components

$\frac{A}{\sin a} = \frac{B}{\sin b} = \frac{C}{\sin c}$ – sine law for the velocity triangle (a opposite A)

$C^2 = A^2 + B^2 - 2AB \cos c$ – cosine law; use with two sides and the included angle

Kinetics: Equations of Motion (Ch. 13)

$\Sigma \mathbf{F} = m\mathbf{a}$ – draw a free-body diagram AND a kinetic ($m\mathbf{a}$) diagram before writing scalar equations

$F_k = \mu_k N$ – kinetic friction; opposes the relative sliding direction

$F_f \leq \mu_s N$ – static friction; equality only at impending slip

$F_s = ks$ – spring force; s = stretch or compression from unstretched length, acts to restore

Rectangular components – best when path/forces are naturally in x, y, z (incline, drag/thrust problems):

$$\Sigma F_x = ma_x, \quad \Sigma F_y = ma_y$$

$$\Sigma F_z = ma_z$$

Normal-tangential components – use when ρ is known (circular motion, banked curves, vertical loops):

$$\Sigma F_t = ma_t = m \frac{dv}{dt}$$

$$\Sigma F_n = ma_n = m \frac{v^2}{\rho}$$

Polar/cylindrical components – use when r, θ are the natural coordinates (e.g. rotating slotted arm):

$$\Sigma F_r = m(\ddot{r} - r\dot{\theta}^2)$$

$$\Sigma F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

$$\Sigma F_z = m\ddot{z}$$

$\frac{d^2 u}{d\theta^2} + u = -\frac{F}{mh^2 u^2}$, $u = \frac{1}{r}$ – central-force/orbital motion; $F > 0$ away from force center O

$h = r^2 \dot{\theta} = \text{const}$ – angular momentum per unit mass, constant for any central force

Kinetics: Work & Energy (Ch. 14)

$U_{1-2} = \int_{s_1}^{s_2} F \cos \theta ds$ – positive when F has a component along motion

$U_W = -W\Delta y - \Delta y$ positive upward

$U_s = -(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2) - s$ from unstretched length; negative when stretching

$T_1 + \Sigma U_{1-2} = T_2$ – principle of work and energy (scalar); include friction as negative work

$T = \frac{1}{2}mv^2$ – kinetic energy of a particle

$P = \frac{dU}{dt} = \mathbf{F} \cdot \mathbf{v}$ – power, in watts (J/s)

$P = Fv$ – force along the direction of motion

$P_{in} = \frac{P_{out}}{\varepsilon}$ – power a motor must draw to deliver P_{out}

$\varepsilon = \frac{P_{out}}{P_{in}} = \frac{\text{energy output}}{\text{energy input}}$ – efficiency; always < 1 (friction, heat losses)

$V_g = Wy$ – gravitational potential energy, measured from a chosen datum (+ above, – below)

$V_e = \frac{1}{2}ks^2$ – elastic potential energy; always ≥ 0 (s = deformation from unstretched length)

$T_1 + V_1 = T_2 + V_2$ – conservation of mechanical energy; valid only if all forces doing work are conservative

Kinetics: Impulse & Momentum (Ch. 15)

$m\mathbf{v}_1 + \Sigma \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2$ – linear impulse-momentum; useful when $F = f(t)$

$\Sigma m_A(v_A)_1 + \Sigma m_B(v_B)_1 = \Sigma m_A(v_A)_2 + \Sigma m_B(v_B)_2$ – conservation, e.g. collisions

$F_{avg} = \frac{m(v_2 - v_1)}{\Delta t}$ – average impulsive force over contact time Δt

$v = \frac{m_A v_A + m_B v_B}{m_A + m_B}$ – common velocity after a plastic impact / coupling ($e = 0$)

$e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$ – $e = 1$ perfectly elastic (KE conserved), $e = 0$ perfectly plastic (move together)

Oblique impact: resolve along line of impact (n) and plane of contact (t); momentum & e apply along n , tangential v unchanged (smooth surfaces)

$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v}$ – angular momentum of a particle about O

$H_O = mvd$ – d = perpendicular distance from O to the line of action of mv

$H_O = mr^2 \dot{\theta} = mrv_\theta$ – angular momentum in polar coordinates

$(H_O)_1 + \Sigma \int_{t_1}^{t_2} M_O dt = (H_O)_2$ – angular impulse-momentum; moments summed about the same fixed O

H_O is conserved (constant) whenever $\Sigma M_O = 0$, as in central-force motion

Quick Reference

$\Sigma \mathbf{F} = m\mathbf{a}$ – Newton's second law; master equation for every kinetics problem

$v = v_0 + a_c t$, $s = s_0 + v_0 t + \frac{1}{2}a_c t^2$, $v^2 = v_0^2 + 2a_c(s - s_0)$ – constant acceleration

$a_t = v dv/ds$, $a_n = v^2/\rho$ – n - t acceleration components

$a_r = \ddot{r} - r\dot{\theta}^2$, $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$ – polar acceleration components

$T_1 + \Sigma U_{1-2} = T_2$ – work-energy principle

$m\mathbf{v}_1 + \Sigma \int \mathbf{F} dt = m\mathbf{v}_2$ – linear impulse-momentum

$e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$ – coefficient of restitution

Rigid-Body Planar Kinematics (Ch. 16)

Types of motion: translation (rectilinear/curvilinear, no rotation)
– rotation about a fixed axis – general plane motion (translation + rotation)

$\omega = \frac{d\theta}{dt}$, $\alpha = \frac{d\omega}{dt} = \omega \frac{d\omega}{d\theta}$ – angular velocity & acceleration (fixed-axis rotation)

$\omega = \omega_0 + \alpha_c t$, $\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha_c t^2$, $\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$ – constant α_c

$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ – velocity of a point on a body rotating about a fixed axis, $\mathbf{r}$ from the axis

$a_t = \alpha r$, $a_n = \omega^2 r$ – tangential & normal acceleration of that point

$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$ – relative velocity, translating axes; A, B on the same rigid body

$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A}$ – relative acceleration, translating axes

Instantaneous center (IC) of zero velocity:

$v_P = \omega r_{P/IC}$ – velocity of any point P , $\perp$ to the line from the IC, proportional to distance from it

Find the IC from perpendiculars to two known velocity directions. The IC generally has nonzero acceleration – never use it for accelerations.

Relative motion, rotating axes (Coriolis): $\boldsymbol{\Omega}$ = angular velocity of the rotating xyz frame (not necessarily the body's ω)

$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} + (\mathbf{v}_{B/A})_{xyz}$

$\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{B/A}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz} + (\mathbf{a}_{B/A})_{xyz}$ – the $2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz}$ term is the Coriolis acceleration; nonzero whenever there's relative motion within the rotating frame

Rigid-Body Kinetics: Force & Acceleration (Ch. 17)

$I = \int r^2 dm$, $k = \sqrt{I/m}$ – mass moment of inertia & radius of gyration; r = distance to the axis

$I = I_G + md^2$ – parallel-axis theorem; d = distance between the parallel axes. Sum $(I_G + md^2)$ for composite bodies, subtract for holes.

Moment of inertia – common shapes:

$I_G = \frac{1}{12} mL^2$ – slender rod, about center, $\perp$ to length L

$I_O = \frac{1}{3} mL^2$ – slender rod, about one end

$I_G = \frac{1}{2} mr^2$ – thin disk or solid cylinder, about central axis

$I_G = mr^2$ – thin ring/hoop, about central axis $\perp$ to plane

$I_G = \frac{2}{5} mr^2$ – solid sphere, about a diameter

$I_G = \frac{1}{12} m(a^2 + b^2)$ – rectangular plate, centroidal axis $\perp$ to plate, sides a, b

Equations of motion:

$\Sigma F_x = m(a_G)_x$, $\Sigma F_y = m(a_G)_y$, $\Sigma M_G = 0$ – pure translation (no rotation)

$\Sigma M_O = I_O \alpha$, $\Sigma F_n = m\omega^2 r_G$, $\Sigma F_t = m\alpha r_G$ – fixed-axis rotation, O on the axis; $I_O = I_G + mr_G^2$

$\Sigma F_x = m(a_G)_x$, $\Sigma F_y = m(a_G)_y$, $\Sigma M_G = I_G \alpha$ – general plane motion (3 independent scalar eqns)

Moments may also be summed about any point P : $\Sigma M_P = \Sigma \mathcal{M}_k$, which must include the moment of $m\mathbf{a}_G$.

Rigid-Body Work & Energy (Ch. 18)

$T = \frac{1}{2} mv_G^2$ – kinetic energy, translation

$T = \frac{1}{2} I_O \omega^2$ – kinetic energy, fixed-axis rotation about O

$T = \frac{1}{2} mv_G^2 + \frac{1}{2} I_G \omega^2$ – kinetic energy, general plane motion

$U_F = \int F \cos \theta ds$ – work of a force

$U_M = \int M d\theta = M\Delta\theta$ (if M constant) – work of a couple moment

Work of weight/spring as in Ch. 14: $U_W = -W\Delta y_G$, $U_s = -\Delta(\frac{1}{2}ks^2)$, using the motion of G /spring end

$T_1 + \Sigma U_{1-2} = T_2$ – principle of work and energy, whole body or connected system; frictionless pins do no net work

$T_1 + V_1 = T_2 + V_2$ – conservation of energy; $V = V_g + V_e$ as in Ch. 14, valid only if all forces doing work are conservative

Rigid-Body Impulse & Momentum (Ch. 19)

$\mathbf{L} = m\mathbf{v}_G$ – linear momentum, acts through G

$H_G = I_G \omega$, $H_O = I_O \omega$ (fixed axis at O) – angular momentum

$H_P = I_G \omega + mv_G d$ – angular momentum about an arbitrary point P ; d = perpendicular distance from P to the line of action of $m\mathbf{v}_G$

$m(\mathbf{v}_G)_1 + \Sigma \int_{t_1}^{t_2} \mathbf{F} dt = m(\mathbf{v}_G)_2$ – linear impulse-momentum, same structure as a particle but using G

$(I_G \omega)_1 + \Sigma \int_{t_1}^{t_2} M_G dt = (I_G \omega)_2$ – angular impulse-momentum; can also use I_O about a fixed point, or the IC in special cases

Conservation: linear momentum conserved with no net external impulse; angular momentum conserved about a point/axis with no net external angular impulse (e.g. impact at a pin)

Eccentric impact: combine conservation of angular momentum about the fixed point/pin (impulsive pin reactions have zero moment there) with the restitution equation $e = [(v_B)_2 - (v_A)_2] / [(v_A)_1 - (v_B)_1]$ applied to the contact-point velocities along the line of impact

Quick Reference — rigid body

$I = I_G + md^2$ – parallel-axis theorem

$\Sigma M_G = I_G \alpha$ – general plane motion (with $\Sigma F_x = ma_{Gx}$, $\Sigma F_y = ma_{Gy}$)

$T = \frac{1}{2} mv_G^2 + \frac{1}{2} I_G \omega^2$ – kinetic energy, general plane motion

$T_1 + \Sigma U_{1-2} = T_2$ – work-energy; $T_1 + V_1 = T_2 + V_2$ if conservative

$m(\mathbf{v}_G)_1 + \Sigma \int \mathbf{F} dt = m(\mathbf{v}_G)_2$, $(I_G \omega)_1 + \Sigma \int M_G dt = (I_G \omega)_2$

$v_P = \omega r_{P/IC}$ – instantaneous center method (velocities only, never accelerations)